CBSE CLASS 10 | MATHEMATICS

PROBABILITY

Master Concept & Formula Handbook

Academic Year 2026–27

Understand the idea. Count carefully.
Calculate with confidence.

Definitions | Key formulae | Worked examples | Common mistakes

A student-friendly Maths Manthan revision resource



What this handbook includes

- Classical probability and complementary events
- Worked examples using coins, cards, dice and counting
- Quick checks to avoid common mistakes

How to use it

Learn the core definition first. Then use the worked examples to practise counting outcomes and simplifying fractions. Extension pages are clearly labelled.

Scope note

The CBSE Class X Mathematics curriculum for 2026–27 lists the classical definition of probability and simple probability problems (see the official CBSE curriculum). This handbook also includes additional practice and enrichment examples; these are labelled **PRACTICE** or **EXTENSION** and should not be read as a guarantee that every topic is prescribed for the board examination. Check your school's current syllabus and prescribed textbook.

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Prepared for learning and revision – not an official CBSE publication.

CHAPTER 01

CORE REVISION

Foundations of Probability

The definition, notation and rules you should know first.

1.1 Random experiments and outcomes

A **random experiment** is an experiment whose possible outcomes are known, but the result of a particular trial cannot be predicted with certainty in advance. Examples include tossing a fair coin, rolling a fair die or drawing one card from a well-shuffled deck.

Term	Meaning
Trial	Performing the experiment once.
Outcome	The result observed in one trial, such as rolling a 4.
Sample space S	The set of all possible outcomes. Its number of elements is $n(S)$.
Event E	A subset of the sample space. The number of outcomes in it is $n(E)$.
Equally likely	Outcomes that have the same probability under the stated model.

1.2 Classical definition of probability

If the elementary outcomes are equally likely, then

$$\boxed{\mathbb{P}(E) = \frac{n(E)}{n(S)} = \frac{\text{number of favourable outcomes}}{\text{total number of equally likely outcomes}}}$$

Here, $n(E)$ counts outcomes that make the event happen, and $n(S)$ counts all possible outcomes.

Example: rolling one fair die

The sample space is $S = \{1, 2, 3, 4, 5, 6\}$. For the event $E =$ "an even number", the favourable outcomes are $\{2, 4, 6\}$. Therefore,

$$\mathbb{P}(E) = \frac{3}{6} = \boxed{\frac{1}{2}}.$$

1.3 Essential probability properties

$$\boxed{0 \le \mathbb{P}(E) \le 1} \qquad \boxed{\mathbb{P}(\overline{E}) = 1 - \mathbb{P}(E)}$$

An impossible event has probability 0; a sure event has probability 1. The event $\overline{E}$ means "E does not occur".

Precision check

The probability must lie between 0 and 1, inclusive. Values such as -0.4, $\frac{5}{3}$ and 125% are invalid probabilities. But $\frac{3}{5} = 0.6$ is valid.

CHAPTER 02

PRACTICE

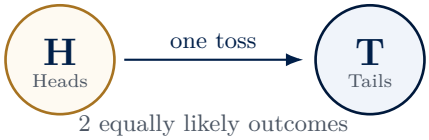
Coin Tossing

Binary outcomes, repeated tosses and "at least / at most".

2.1 One fair coin

A fair coin has two modelled outcomes: Heads (H) and Tails (T). Under the fair-coin assumption,

$$\mathbb{P}(H) = \frac{1}{2}, \qquad \mathbb{P}(T) = \frac{1}{2}.$$



2.2 Number of outcomes for repeated tosses

If a fair coin is tossed n times and each toss has two possible outcomes, then the total number of ordered outcome sequences is

$$\boxed{n(S) = 2^n}$$

Examples: one toss gives 2 outcomes; two tosses give 4; three tosses give 8.

Tosses	Sample space	Total
1	$\{H, T\}$	$2^1 = 2$
2	$\{HH, HT, TH, TT\}$	$2^2 = 4$
3	$\{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$	$2^3 = 8$

2.3 Useful three-toss examples

Event	Probability	Event	Probability
All heads	1/8	All tails	1/8
Exactly 2 heads	3/8	Exactly 1 head	3/8
At least 2 heads	$4/8 = 1/2$	At most 2 heads	7/8

Shortcut: use the complement

"At least one head" is easier to calculate as the complement of "no heads":

$$\mathbb{P}(\text{at least one head}) = 1 - \mathbb{P}(\text{all tails}) = 1 - \left(\frac{1}{2}\right)^n.$$

Language decoder

At least k means k or more. **At most** k means k or fewer, including zero.

CHAPTER 03

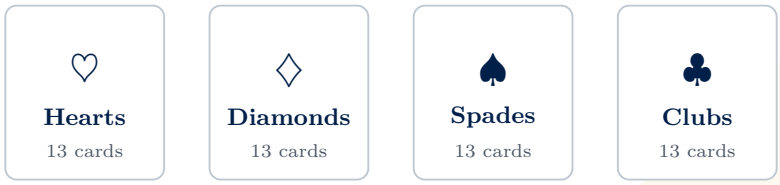
PRACTICE

Playing Cards: The 52-Card Deck

Count suits and card types before applying the probability formula.

3.1 How a standard deck is organised

A standard deck used in textbook probability contains 52 cards; jokers are excluded. It has four suits, with 13 cards in each suit.



Red suits

Hearts: 13 cards
Diamonds: 13 cards
Total red cards: $13 + 13 = 26$

Black suits

Spades: 13 cards
Clubs: 13 cards
Total black cards: $13 + 13 = 26$

$$\mathbb{P}(\text{red}) = \frac{26}{52} = \frac{1}{2}, \qquad \mathbb{P}(\text{black}) = \frac{26}{52} = \frac{1}{2}.$$

3.2 Card categories worth remembering

Category	What it contains	Count
Face cards	Kings, Queens and Jacks; 3 in each suit	12
Aces	One Ace per suit; Aces are not face cards	4
Honor cards	A, K, Q and J; 4 in each suit	16
Number cards	2 through 10; 9 in each suit	36
Non-face cards	Everything except K, Q and J; includes Aces	40

Common card-counting mistake

An Ace is not a face card. Face cards are only the Kings, Queens and Jacks: $3 \times 4 = 12$ cards. Therefore, the number of non-face cards is $52 - 12 = 40$.

PRACTICE

Card Reference Table

Use the deck counts to build the numerator and denominator correctly.

3.3 At-a-glance reference table

Card category	♡	♢	♠	♣	Total	Probability
Face cards (K, Q, J)	3	3	3	3	12	$12/52 = 3/13$
Kings	1	1	1	1	4	$4/52 = 1/13$
Queens	1	1	1	1	4	$4/52 = 1/13$
Jacks	1	1	1	1	4	$4/52 = 1/13$
Aces	1	1	1	1	4	$4/52 = 1/13$
Honor cards (A, K, Q, J)	4	4	4	4	16	$16/52 = 4/13$
Number cards (2–10)	9	9	9	9	36	$36/52 = 9/13$
Non-face cards	10	10	10	10	40	$40/52 = 10/13$
Red cards	13	13	0	0	26	$1/2$
Black cards	0	0	13	13	26	$1/2$

3.4 When cards are removed

If k cards are removed from the deck before a card is drawn, the number of remaining cards is $52 - k$. Count the remaining favourable cards as well.

$$\boxed{\text{new total} = 52 - k} \qquad \boxed{\mathbb{P}(E) = \frac{\text{favourable cards remaining}}{\text{cards remaining}}}$$

Worked example

Question. All six red face cards are removed from a standard deck. Find the probability of drawing (i) a king, (ii) a red card, and (iii) a spade.

The six removed cards are $2K + 2Q + 2J = 6$, so $52 - 6 = 46$ cards remain.

(i) Two black kings remain: $\mathbb{P}(\text{king}) = \dfrac{2}{46} = \boxed{\dfrac{1}{23}}$.

(ii) Red cards remaining: $26 - 6 = 20$, so $\mathbb{P}(\text{red}) = \dfrac{20}{46} = \boxed{\dfrac{10}{23}}$.

(iii) No spades were removed: $\mathbb{P}(\text{spade}) = \boxed{\dfrac{13}{46}}$.

Denominator check

After removing cards, do not continue using 52 as the denominator. Both the total deck size and the favourable count must reflect what remains.

CHAPTER 04

PRACTICE

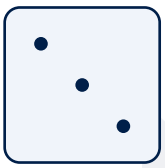
A Single Fair Die

List the sample space, then count the outcomes that satisfy the event.

4.1 Sample space

A standard fair die has six possible outcomes when rolled once:

$$S = \{1, 2, 3, 4, 5, 6\}, \qquad n(S) = 6, \qquad \mathbb{P}(\text{each individual outcome}) = \frac{1}{6}.$$



A fair die is modelled as having six equally likely outcomes.

$$S = \{1, 2, 3, 4, 5, 6\}$$

4.2 Common events

Event	Favourable outcomes	Probability
Even number	$\{2, 4, 6\}$	$3/6 = 1/2$
Odd number	$\{1, 3, 5\}$	$3/6 = 1/2$
Prime number	$\{2, 3, 5\}$	$3/6 = 1/2$
Composite number	$\{4, 6\}$	$2/6 = 1/3$
Perfect square	$\{1, 4\}$	$2/6 = 1/3$
Number greater than 4	$\{5, 6\}$	$2/6 = 1/3$
Multiple of 2 or 3	$\{2, 3, 4, 6\}$	$4/6 = 2/3$

Example

Let E be the event "rolling a prime number". On a die, $E = \{2, 3, 5\}$, so

$$\mathbb{P}(E) = \frac{n(E)}{n(S)} = \frac{3}{6} = \boxed{\frac{1}{2}}.$$

Do not forget

The number 1 is neither prime nor composite. The smallest prime number is 2; it is also the only even prime.

CHAPTER 05

EXTENSION

Two Dice: The 36-Outcome Model

For two distinguishable dice, outcomes are ordered pairs.

5.1 The sample space table

If Die 1 shows x and Die 2 shows y, each outcome is (x, y). There are $6 \times 6 = 36$ equally likely ordered pairs when both dice are fair and independent.

$$\boxed{n(S) = 6 \times 6 = 36}$$

Die 1 \ Die 2	1	2	3	4	5	6
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

Ordered pairs are important

$(2, 5)$ and $(5, 2)$ are different outcomes: the first entry refers to Die 1, and the second to Die 2.

5.2 Quick results

Event	Probability	Event	Probability
Both dice match (a doublet)	$6/36 = 1/6$	Not a doublet	$30/36 = 5/6$
First die $>$ second die	$15/36 = 5/12$	First die $<$ second die	$15/36 = 5/12$
Product is odd	$9/36 = 1/4$	Product is even	$27/36 = 3/4$
Neither die shows 5	$25/36$	At least one die shows 5	$11/36$

Why are there 15 pairs with first die greater?

The 6 diagonal pairs have equal values. The remaining $36 - 6 = 30$ pairs split symmetrically into 15 with the first value larger and 15 with the first value smaller.

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Two-Dice Sums

The middle sum is more likely because more pairs produce it.

5.3 Sum frequency table

The sum $s = x + y$ ranges from 2 to 12. The probability of a sum equals the number of ordered pairs producing that sum divided by 36.

Sum s	2	3	4	5	6	7	8	9	10	11	12
Favourable pairs	1	2	3	4	5	**6**	5	4	3	2	1
Probability	1/36	2/36	3/36	4/36	5/36	**1/6**	5/36	4/36	3/36	2/36	1/36

Shortcut for counting pairs

- For $2 \le s \le 7$, favourable pairs $= s - 1$.
- For $8 \le s \le 12$, favourable pairs $= 13 - s$.

Example: for sum 5 there are $5 - 1 = 4$ pairs, so $\mathbb{P}(\text{sum } 5) = 4/36 = 1/9$.

5.4 More event examples

Event	Counting	Probability
Prime sum	Sums 2, 3, 5, 7, 11: $1 + 2 + 4 + 6 + 2 = 15$ pairs	$15/36 = 5/12$
Composite sum	Complement of prime sums; 21 pairs	$21/36 = 7/12$
Perfect-square sum	Sums 4 and 9: $3 + 4 = 7$ pairs	$7/36$
Sum is a multiple of 3	Sums 3, 6, 9, 12: $2 + 5 + 4 + 1 = 12$	$12/36 = 1/3$
Sum at least 10	Sums 10, 11, 12: $3 + 2 + 1 = 6$	$6/36 = 1/6$
Sum at most 5	Sums 2, 3, 4, 5: $1 + 2 + 3 + 4 = 10$	$10/36 = 5/18$

Check the model

These results assume two fair, independent dice and count ordered pairs. If the experiment is defined differently, the sample space may also be different.

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CHAPTER 06

EXTENSION

Bags, Unknown Counts and Numbered Items

Set up the total count before writing a probability fraction.

6.1 Sampling from a bag

If a bag has R red items out of N items in total, and each item is equally likely to be drawn, then

$$\mathbb{P}(\text{red}) = \frac{R}{N}, \qquad \mathbb{P}(\text{not red}) = 1 - \frac{R}{N} = \frac{N-R}{N}.$$

Worked example: find an unknown count

A bag contains 5 red balls and x blue balls. The probability of drawing a blue ball is three times the probability of drawing a red ball. Find x.

$$\frac{x}{5+x} = 3\left(\frac{5}{5+x}\right) \implies x = 15.$$

Worked example: adding objects

A bag has x red balls among 12 balls. Six additional red balls are added, and the probability of drawing red doubles. Find x.

$$\frac{x+6}{18} = 2\left(\frac{x}{12}\right) \implies \frac{x+6}{18} = \frac{x}{6} \implies x+6 = 3x \implies \boxed{x = 3}.$$

6.2 With or without replacement?

With replacement	The selected item is returned before the next draw. The total count remains the same.
Without replacement	The selected item is not returned. The total count decreases by one after each draw.

6.3 Counting consecutive numbers

If items are numbered from a to b, inclusive, then

$$\boxed{\text{number of integers from } a \text{ to } b = b - a + 1}$$

From 10 to 90: $90 - 10 + 1 = 81$ integers. From 1 to 100: 100 integers.

6.4 Counting multiples

The number of multiples of a positive integer k among $1, 2, \ldots, N$ is $\lfloor N/k \rfloor$.

- Multiples of 5 from 1 to 90: $\lfloor 90/5 \rfloor = 18$, so the probability is $18/90 = 1/5$.
- Numbers divisible by both 2 and 3 are multiples of $\text{lcm}(2,3) = 6$. From 1 to 100, there are $\lfloor 100/6 \rfloor = 16$.

Inclusive endpoints

The "+1" matters because both endpoints are included. The interval from 10 through 90 contains 81 integers, not 80.

CHAPTER 07

EXTENSION

Calendar Probability

An enrichment topic: count the extra days after complete weeks.

7.1 Ordinary year and leap year

Ordinary year
365 days $= 52 \times 7 + 1$
52 weeks + 1 extra day

Leap year
366 days $= 52 \times 7 + 2$
52 weeks + 2 extra days

A Gregorian leap year is divisible by 4, except century years must also be divisible by 400. For example, 2000 was a leap year; 1900 was not.

7.2 Probability of extra Sundays

Assuming each possible weekday alignment is equally likely:

Question	Reason	Probability
53 Sundays in an ordinary year	One extra day can be Sunday	1/7
53 Sundays in a leap year	Two consecutive extra days can include Sunday in two alignments	2/7
53 Sundays and 53 Mondays in a leap year	The two extra days must be Sunday and Monday	1/7
53 Sundays or 53 Mondays in a leap year	Three alignments satisfy at least one event	3/7

7.3 February and birthdays

- February in an ordinary year has 28 days, exactly four weeks. It has exactly four Sundays, Mondays, and so on.
- February in a leap year has 29 days: four weeks plus one extra day. Under the equal-weekday assumption, the probability of five Sundays is 1/7.
- For two independently selected birthdays in a 365-day model, the probability of the same birthday is 1/365, and of different birthdays is 364/365.

Model assumptions matter

Calendar probabilities use an idealised model in which weekday alignments are treated as equally likely. Birthday examples commonly ignore leap day and assume independent, uniformly distributed birthdays.

CHAPTER 08

EXTENSION

Geometric Probability

Enrichment: use length, area or angle only when the model justifies it.

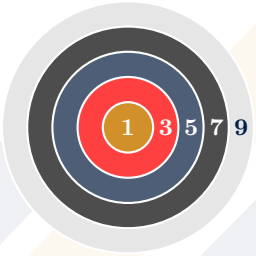
8.1 The measure-ratio idea

If a point is selected uniformly from a region, probability can be modelled as the measure of the favourable region divided by the measure of the total region.

$$\mathbb{P}(E) = \frac{\text{length, area or volume of favourable region}}{\text{length, area or volume of total region}}$$

8.2 Concentric circles and annular areas

For circles with radii $r, 2r, 3r, 4r, 5r$, the five band areas are proportional to the consecutive odd numbers $1 : 3 : 5 : 7 : 9$.



$$A_1 = \pi r^2, \quad A_2 = \pi(2r)^2 - \pi r^2 = 3\pi r^2,$$
$$A_3 = 5\pi r^2, \quad A_4 = 7\pi r^2, \quad A_5 = 9\pi r^2.$$

The total target area is $\pi(5r)^2 = 25\pi r^2$. Therefore, the probability of landing in the innermost region is $1/25$; the second band is $3/25$; and the third band is $5/25 = 1/5$.

8.3 Circle inside a square

A circle of radius r inscribed in a square has side $2r$. The area of the square is $4r^2$, while the circle's area is πr^2.

$$\mathbb{P}(\text{inside circle}) = \frac{\pi r^2}{4r^2} = \frac{\pi}{4} \approx 0.7854.$$

$$\mathbb{P}(\text{outside circle but inside square}) = 1 - \frac{\pi}{4} = \frac{4-\pi}{4} \approx 0.2146.$$

8.4 Spinner sectors

For a spinner divided into sectors of central angle θ_i, with a uniformly distributed stopping direction,

$$\mathbb{P}(\text{sector } i) = \frac{\theta_i}{360^\circ}.$$

Do not overgeneralise

A ratio of areas or angles is valid only when the model makes the point or direction uniform over the relevant region. This is an enrichment model, not a replacement for the classical definition used in the core chapter.

CHAPTER 09

CORE REVISION

10 Common Probability Mistakes

A final checklist for clearer, more accurate solutions.

1. **Confusing "at least" and "at most".** At least k means $\geq k$; at most k means $\leq k$, including zero.
2. **Forgetting the complement.** $\mathbb{P}(\overline{E}) = 1 - \mathbb{P}(E)$. This can make a count much shorter.
3. **Ignoring overlap.** When using the union rule, account for outcomes in both events: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
4. **Miscounting an inclusive range.** From a through b contains $b - a + 1$ integers.
5. **Calling 1 prime or composite.** The number 1 is neither; 2 is the smallest prime.
6. **Treating an Ace as a face card.** Face cards are Kings, Queens and Jacks.
7. **Keeping an old denominator after removal.** Update the total number of possible outcomes after any removal.
8. **Treating ordered pairs as identical.** For two distinguishable dice, $(2, 5) \neq (5, 2)$.
9. **Accepting a probability outside $[0, 1]$.** Values such as $5/3$, -0.4 and 125% are invalid; $3/5$ is valid.
10. **Forgetting assumptions.** The classical formula $n(E)/n(S)$ requires equally likely elementary outcomes. State the model when needed.

A three-step board-style presentation

Step 1. Define the sample space. State the total number of equally likely outcomes.

Step 2. Count favourable outcomes. List them when doing so makes the reasoning clearer.

Step 3. Calculate and simplify. Write the ratio and reduce it to the lowest terms.

Final self-check

Before moving on, ask yourself:

- Did I count all possible outcomes correctly?
- Did I count only the outcomes that satisfy the event?
- Did I check the assumptions and simplify the fraction?
- Is my answer between 0 and 1, inclusive?

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Keep practising. Keep checking. Keep improving.

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